

A

From a task error to a local update



$$\epsilon = A \Gamma c$$

A: address
which sites

Γ : gain
how strongly

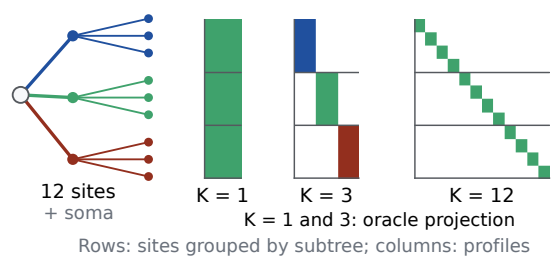
c: coefficient
which signal

Update = -learning rate $\times$ local eligibility $\times$ credit

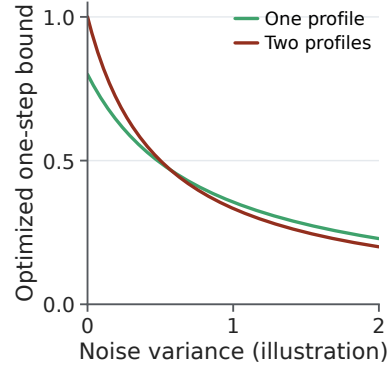
Error source and spatial delivery are separate choices

B

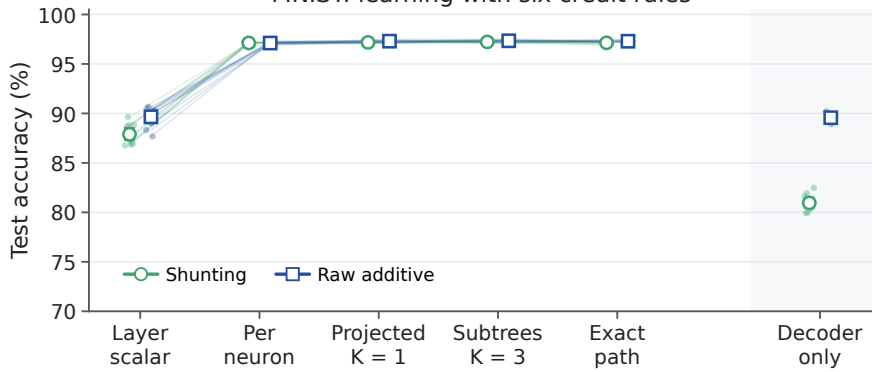
Morphology supplies spatial profiles
Delivered field = $A c$

**C**

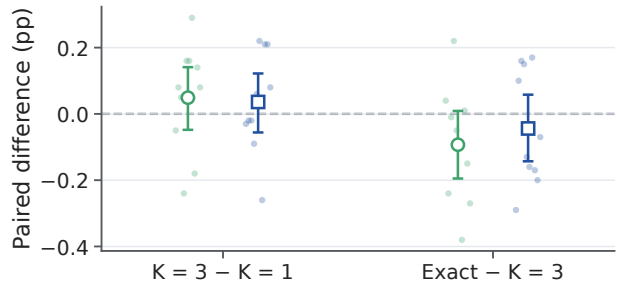
Noise and resolution

**D**

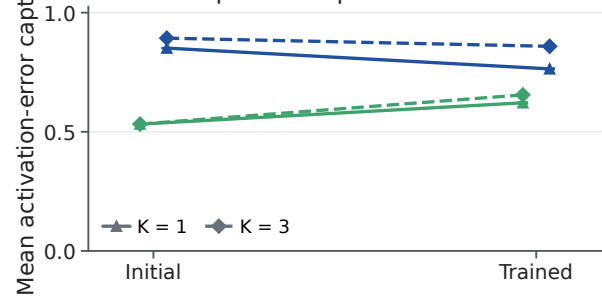
MNIST: learning with six credit rules

**E**

Additional resolution within a neuron

**F**

More profiles capture more credit

**G**

Separate image controls

